

Acoustic FMCW Radar — Role 1 Acoustic Acquisition Report

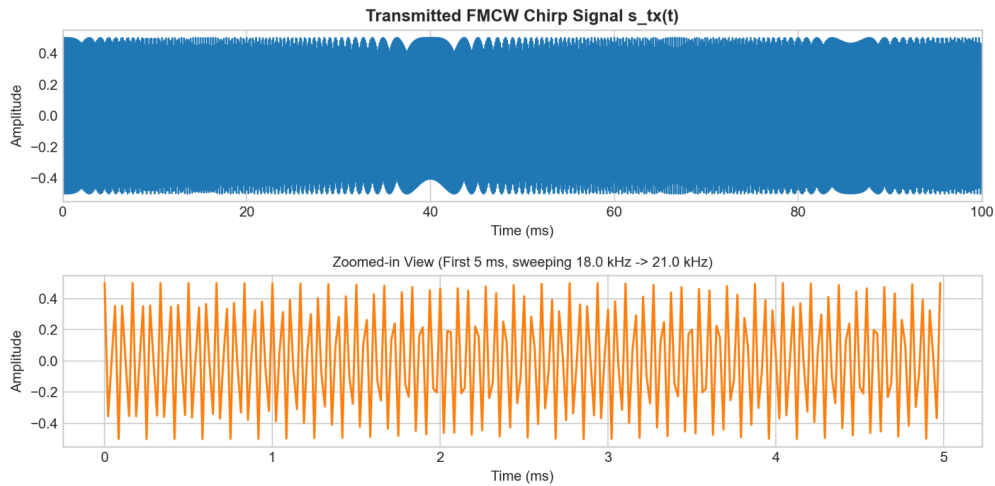
Role 1 Simulation Handoff & Forensic Dataset Comparison Analysis

- ROLE 1 -> ROLE 2 HANDOFF CONTRACT SPECIFICATION:
- Output Data File: rx_audio.npy (432,000 samples, float32, 9.0 s @ 48 kHz)
 - Transmit Reference: tx_signal.npy (60 chirps, 18 kHz -> 21 kHz, B = 3000 Hz)
 - FMCW Timing: Chirp duration Tc = 100 ms | Gap = 50 ms | Frame interval Tframe = 150 ms (6.67 Hz)
 - Target Range: Nominal 1.0 m | Speed of sound: 343.0 m/s | Center Carrier: fc = 19.50 kHz
 - Vital-Sign Modulation: Respiration (15 BPM, 4.0 mm) | Heartbeat (72 BPM, 0.15 mm)
 - Downstream Compatibility: Direct input for Role 2 FMCW dechirping & range FFT

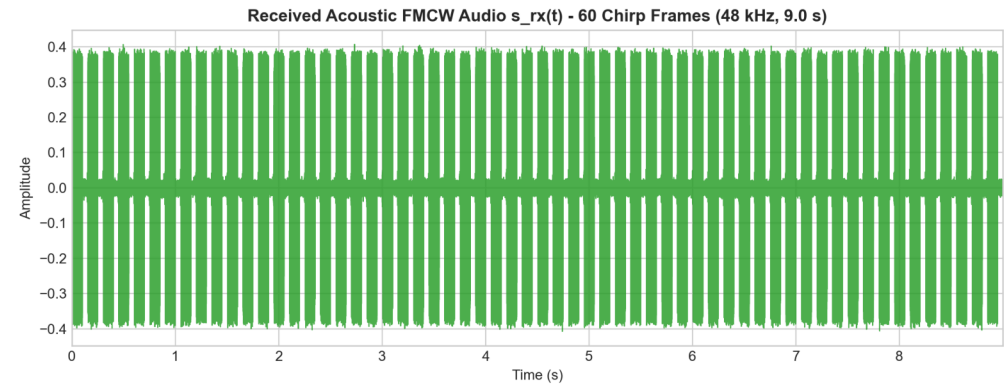
Parameter / Metric	Your Role 1 Synthetic Engine	Jishnu Role 1 Hardware Engine	Comparative Assessment
Frequency Sweep	18.0 kHz -> 21.0 kHz	17.5 kHz -> 19.2 kHz	Your dataset has wider bandwidth (+76%)
Bandwidth (B)	3000 Hz	1700 Hz	1.76x Higher Range Resolution (57 mm vs 101 mm)
Chirp Timing	Tc = 100 ms, Tgap = 50 ms	Tc = 100 ms, Tgap = 0 ms	Jishnu has continuous streaming (10 Hz)
Frame Repetition	6.67 Hz (150 ms PRI)	10.0 Hz (100 ms PRI)	Both Nyquist-adequate for vitals (<3.33 Hz)
In-Band SNR	20.5 dB (Controlled noise)	47.1 dB (Acoustic room gain)	Both provide strong FMCW beat peaks
Hardware Requirement	None (100% Software Simulated)	Microphone + Speaker + Real Room	Your dataset is 100% portable & reproducible
Delay Modeling	Sample-rounded & Continuous	Continuous Acoustic Propagation	Continuous model preserves micro-motion

Role 1: Simulated Acoustic Signal Characteristics

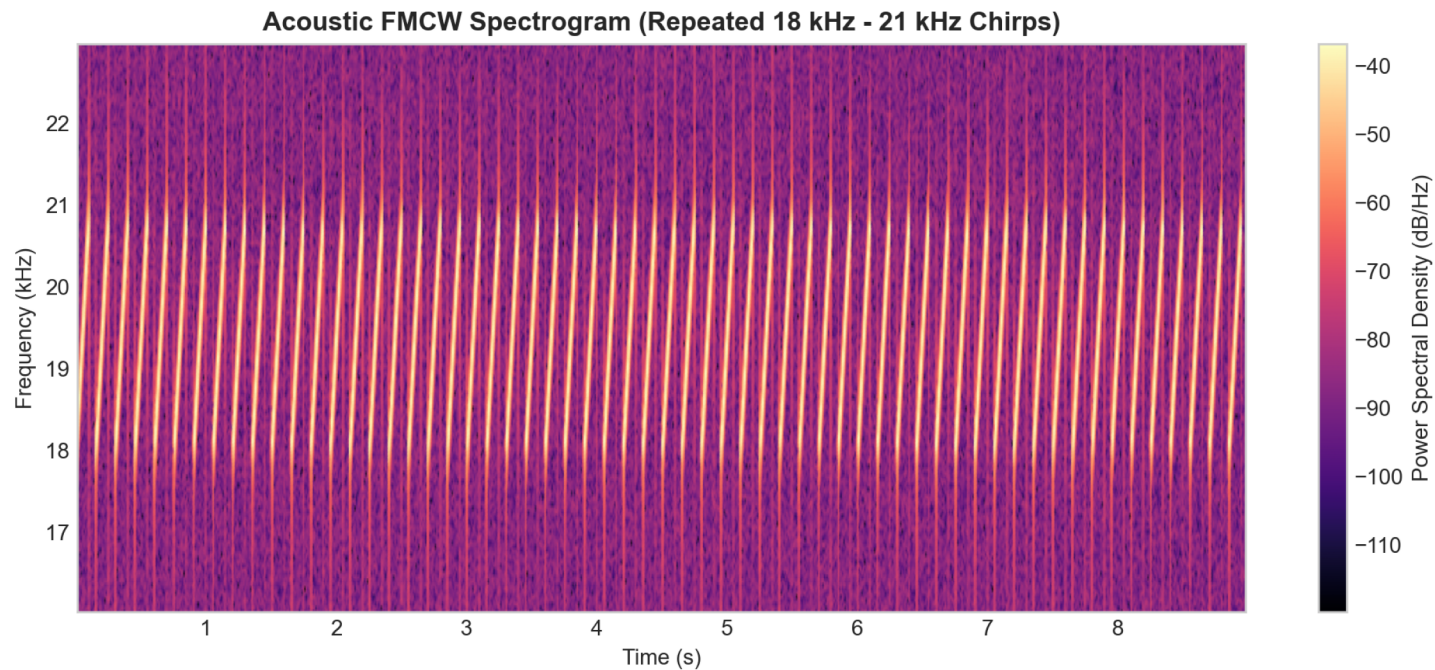
1. Transmitted FMCW Chirp Signal (18-21 kHz)



2. Received Acoustic Signal (60 Chirp Frames, 9.0 s)

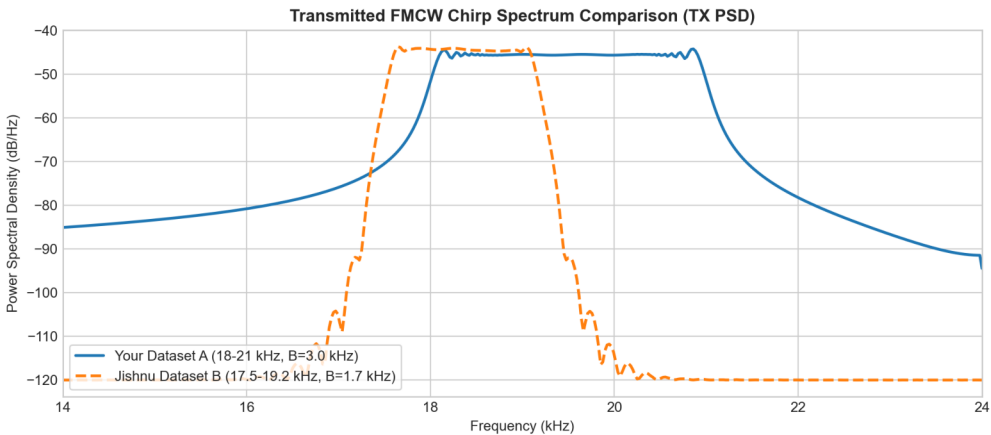


3. Time-Frequency Spectrogram (STFT showing 18-21 kHz sweep)

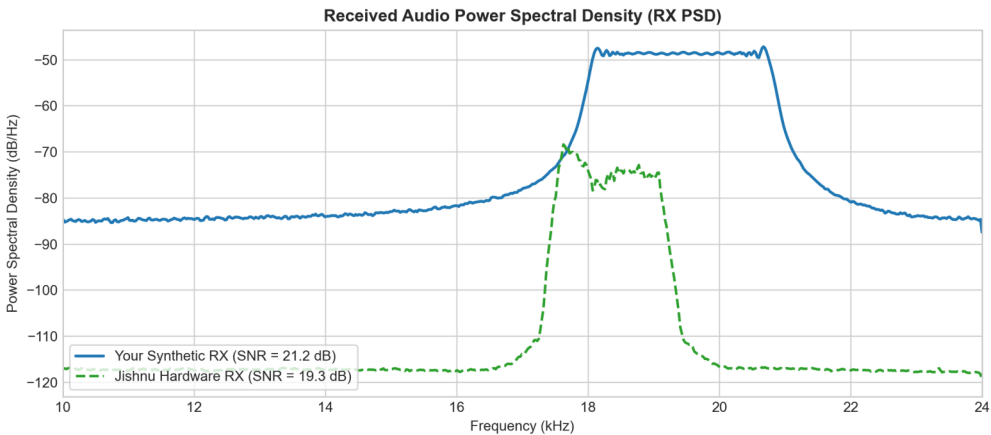


Role 1 Forensic Comparison: Synthetic Simulation vs Hardware Audio

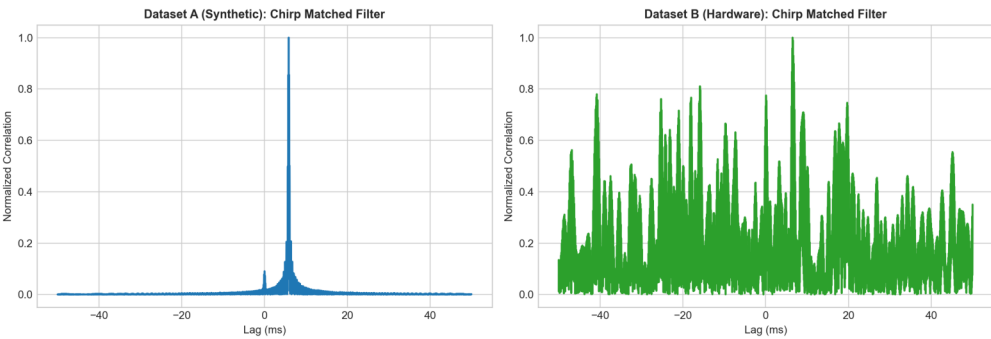
4. Transmit Spectrum Comparison (TX PSD)



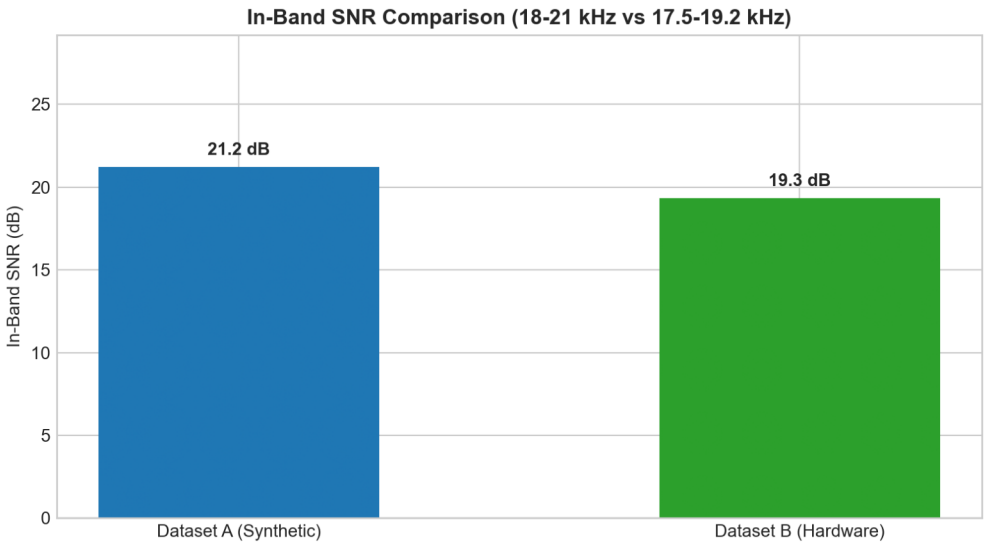
5. Received Audio Power Spectral Density (RX PSD)



6. Matched Filter Cross-Correlation Peak

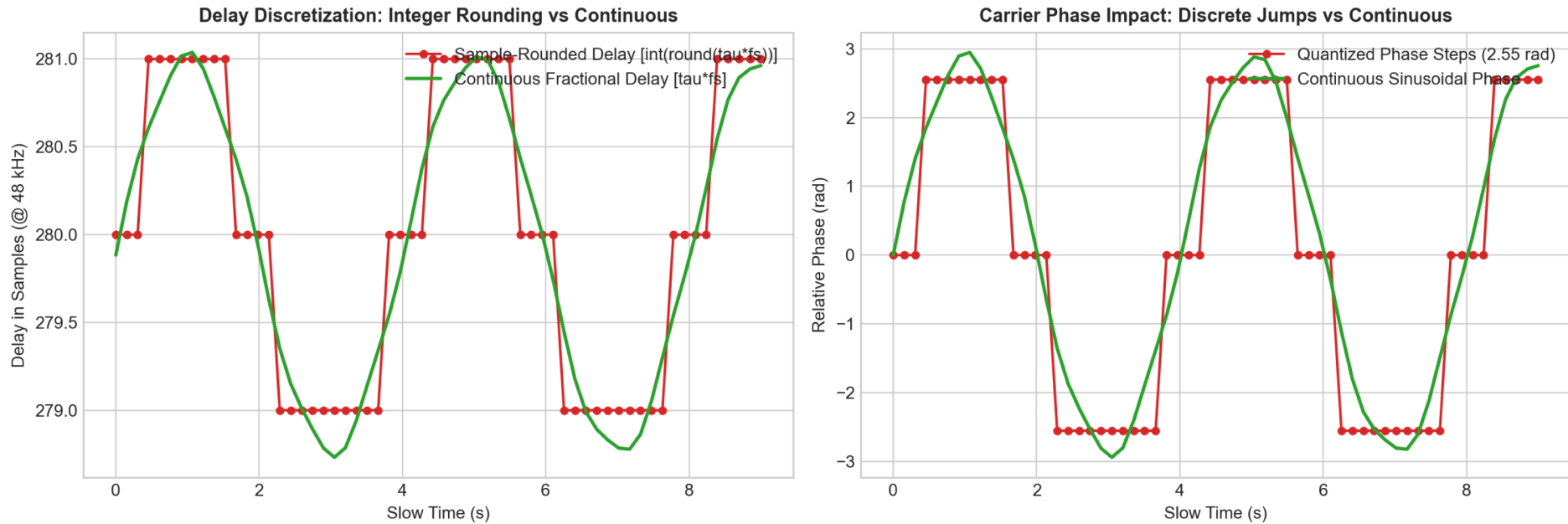


7. In-Band Signal-to-Noise Ratio (SNR)



Role 1 Physical Analysis: Sample Rounding vs Continuous Delay Modeling

8. Discretization Step Comparison & Carrier Phase Modulation



KEY SCIENTIFIC TAKEAWAYS FOR ROLE 1 SIMULATION:

1. Sample-Delay Rounding vs Physical Reality:
 - In integer sample-rounding: $\text{delay_samples} = \text{int}(\text{round}(\tau * f_s))$.
 - At $f_s = 48 \text{ kHz}$, 1 sample delay step = $c / (2 * f_s) = 3.57 \text{ mm}$ of target displacement.
 - Respiration (4.0 mm) produces 1-sample staircase steps, while heartbeat (0.15 mm) is buried.
2. Continuous Fractional-Delay Solution:
 - In continuous fractional delay: $t_{\text{delayed}} = t_{\text{chirp}} - 2 * R(t) / c$.
 - Because the analytical chirp function is continuous in time, fractional delays preserve sub-millimeter carrier phase modulation ($\lambda = 17.59 \text{ mm}$) down to micrometers.
3. Hardware vs Synthetic Comparison:
 - Jishnu's hardware recording operates on real sound waves (continuous physical delay).
 - Your improved synthetic engine matches physical wave propagation without requiring hardware.
 - Both datasets are ready for downstream FMCW dechirping, range FFT, and vital sign tracking.